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ASSIGNMENT II
DEEP LEARNING AND ITS APPLICATIONS — CO3133
DEEP LEARNING ON LARGE SCALE DATA AND SPECIALIZED
TASKS

Deep Learning on Large-Scale Data for Specialized
Computer Vision or Language Tasks

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Contents

1	Dataset Proposal	1
1.1	Dataset Name, Source, Version, and License	1
1.2	Task, Input, and Output Definition	1
1.3	Number of Samples and Annotation Types	2
1.4	Image Resolution Analysis and Preprocessing	2
1.5	Preliminary Distribution Analysis	4
1.6	Data-Splitting Plan	5
1.7	Split Unit to prevent Data Leakage	5
1.8	Evaluation Metrics	5
1.9	Baseline Plan	6
1.10	Planned Pretrained Models	6
1.11	Compute Estimate	7
1.12	Subset Selection Rules	8

List of Figures

List of Tables

1	Dataset Overview	1
2	Task, input, and output specification.	2
3	Dataset size by predefined split.	2
4	Top ten image resolutions and remaining resolutions in the Food-11 training set.	3
5	Training images grouped by the number of dimensions strictly greater than 512 pixels.	3
6	Preliminary class distribution of Food-11 across the predefined splits.	4
7	Planned computational budget for the main Food-11 experiments.	7
8	Measured GPU memory for one forward pass on a batch size 32, 512×512 resolution images	8

1. Dataset Proposal

1.1. Dataset Name, Source, Version, and License

We propose to use the Food-11 image dataset for the image classification task in this project. The dataset contains food images grouped into 11 major food categories: Bread, Dairy product, Dessert, Egg, Fried food, Meat, Noodles/Pasta, Rice, Seafood, Soup, and Vegetable/Fruit.

The Food-11 dataset used in this project is originally provided by the Multimedia Signal Processing Group (MMSPG) at [EPFL](#) through its official Food Image Dataset webpage. The dataset contains 16,643 food images grouped into 11 food categories and is provided with predefined training, validation, and evaluation partitions.

For practical data access and reproducibility, the dataset is obtained through the [Kaggle Food-11 dataset](#) using the Kaggle API. The Kaggle repository is treated only as a convenient distribution channel for accessing the dataset and does not represent the original source of the Food-11 dataset.

Table 1: Dataset Overview

Field	Description
Dataset name	Food-11 image dataset
Source	Kaggle: trolukovich/food11-image-dataset
Original source	Multimedia Signal Processing Group (MMSPG), EPFL
Kaggle publisher	Aleksandr Antonov
Version	Version 1
License	CC0: Public Domain for the Kaggle mirror
Dataset size	16,643 RGB food images
Number of classes	11

1.2. Task, Input, and Output Definition

The proposed machine learning problem is formulated as a multi-class, single-label image classification task.

Given an RGB food image x , the model predicts exactly one class y from the set of 11 Food-11 categories:

$$y \in \{\text{Bread, Dairy product, Dessert, Egg, Fried food, Meat, Noodles/Pasta, Rice, Seafood, Soup, Vegetable/Fruit}\}.$$

Formally, the model learns a mapping

$$f_{\theta} : \mathbb{R}^{H \times W \times 3} \rightarrow \{1, \dots, 11\},$$

where H and W denote the image height and width and the three channels correspond to RGB.

Table 2: Task, input, and output specification.

Component	Definition
Task	11-class food image classification
Input	RGB food image
Output	One categorical label among the 11 Food-11 classes
Annotation type	Image-level categorical class label

1.3. Number of Samples and Annotation Types

The complete Food-11 dataset contains 16,643 images. The predefined dataset split contains 9,866 training images, 3,430 validation images, and 3,347 evaluation/test images.

Table 3: Dataset size by predefined split.

Split	Images	Percentage	Role
Training	9,866	59.28%	Model fitting
Validation	3,430	20.61%	Model selection / early stopping
Evaluation	3,347	20.11%	Final test evaluation
Total	16,643	100%	

The dataset provides only image-level class labels, with each image assigned to a single food category. Therefore, the proposed task is formulated as a single-label, multi-class image classification problem.

1.4. Image Resolution Analysis and Preprocessing

To avoid information leakage from the validation and evaluation partitions into preprocessing decisions, the input resolution and resizing strategy will be determined using only the predefined training partition of Food-11, which contains 9,866 images. The resulting preprocessing rule will be fixed before model training and applied consistently to all three partitions. Table 4 summarizes the ten most frequent training-set resolutions, expressed as width $\times$ height. All remaining resolutions are grouped into Other. Percentages are calculated relative to the total number of training images. The training images have heterogeneous resolutions, but 512×512 is the most common size, accounting for 5,551 images (56.26%). Several other frequent resolutions, including 512×384 and 384×512 , also have a longer dimension of 512 pixels. These observations support selecting 512×512 as the target input size, while preserving the aspect ratio of non-square images.

Table 4: Top ten image resolutions and remaining resolutions in the Food-11 training set.

Rank	Resolution	Images	Percentage
1	512×512	5,551	56.26%
2	512×384	1,355	13.73%
3	384×512	519	5.26%
4	382×512	320	3.24%
5	512×382	216	2.19%
6	512×341	142	1.44%
7	1024×768	108	1.09%
8	512×306	107	1.08%
9	512×289	81	0.82%
10	511×512	60	0.61%
—	Other	1,407	14.26%
Total		9,866	100.00%

Table 5 further groups the training images according to the number of dimensions strictly greater than 512 pixels. The three groups are mutually exclusive: an image with both dimensions greater than 512 belongs only to the two-dimension group. A dimension equal to 512 is not counted as exceeding the threshold.

Table 5: Training images grouped by the number of dimensions strictly greater than 512 pixels.

Dimensions exceeding 512 pixels	Images	Percentage
Neither dimension	9,319	94.46%
Exactly one dimension	120	1.22%
Both dimensions	427	4.33%
Total	9,866	100.00%

Percentages are rounded independently and may not sum to exactly 100%.

Only 547 training images (5.54%) have at least one dimension greater than 512 pixels, whereas 9,319 images (94.46%) have both dimensions at most 512 pixels. Together with the observation that 512×512 is the most frequent training-set resolution (56.26%), this distribution supports selecting 512×512 as the target input size. This choice preserves the native resolution of the largest group of training images and requires downsampling for only a small proportion of the training set, while providing a uniform input size for all models.

However, directly resizing non-square images to 512×512 would alter their original aspect ratios. Therefore, each image will first be resized so that its longer dimension is 512 pixels while preserving its aspect ratio. The shorter dimension will then be padded to 512 pixels, yielding a fixed 512×512 input without cropping the image content or distorting its geometric proportions.

Images already sized 512×512 require neither resizing nor padding. Non-square images whose longer dimension is already 512 require only padding along the shorter dimension. Images with at least one dimension greater than 512 will be downsampled so that their longer dimension becomes 512 pixels.

For images with both dimensions smaller than 512, the image will be upsampled until its longer dimension reaches 512 pixels, while preserving the original aspect ratio and then padded along its shorter dimension to obtain a 512×512 input. Padding is applied only when the resized image is non-square, square images become 512×512 directly after resizing.

1.5. Preliminary Distribution Analysis

Food-11 is not perfectly class balanced. The class distribution reported for the predefined training, validation, and evaluation partitions is shown in Table 6.

Table 6: Preliminary class distribution of Food-11 across the predefined splits.

Class	Train		Val		Test		Total
	N	%	N	%	N	%	
Bread	994	10.08%	362	10.55%	368	10.99%	1,724
Dairy product	429	4.35%	144	4.20%	148	4.42%	721
Dessert	1500	15.20%	500	14.58%	500	14.94%	2,500
Egg	986	9.99%	327	9.53%	335	10.01%	1,648
Fried food	848	8.60%	326	9.50%	287	8.57%	1,461
Meat	1325	13.43%	449	13.09%	432	12.91%	2,206
Noodles/Pasta	440	4.46%	147	4.29%	147	4.39%	734
Rice	280	2.84%	96	2.80%	96	2.87%	472
Seafood	855	8.67%	347	10.12%	303	9.05%	1,505
Soup	1500	15.20%	500	14.58%	500	14.94%	2,500
Vegetable/Fruit	709	7.19%	232	6.76%	231	6.90%	1,172
Total	9866	100%	3430	100%	3347	100%	16643

The preliminary analysis indicates a moderate class imbalance. In the training set, the largest classes are Dessert and Soup (1,500 images each), while Rice is the smallest class with 280 images. Thus, the largest-to-smallest class ratio is approximately $1500/280 \approx 5.36$.

The predefined validation and evaluation partitions follow a similar overall distribution to the training set. Nevertheless, the unequal class frequencies motivate the use of macro-Precision, macro-Recall, macro-F1 as the primary evaluation metrics, so that performance on smaller classes is not completely dominated by the largest classes.

1.6. Data-Splitting Plan

The project will preserve the official three-way split provided by Food-11:

Training → Model fitting, Validation → Model selection, Evaluation → Final testing.

No random re-splitting of the complete dataset is planned because the dataset already provides predefined training, validation, and evaluation partitions.

The validation set will be used for model selection, early stopping, and checkpoint selection. . The test set will only be used after training for the final comparison of models.

1.7. Split Unit to prevent Data Leakage

The split unit is the individual image. The dataset does not provide a subject, user, plate, restaurant, or scene identifier that can be used as a higher-level grouping variable. Therefore, entity-level grouped splitting is not possible with the available metadata.

1.8. Evaluation Metrics

The primary evaluation metrics will be

Macro-F1, Macro-Precision, Macro-Recall

because the dataset is moderately imbalanced across the 11 classes.

For class c , the F1 score is

$$F1_c = 2 \frac{\text{Precision}_c \text{Recall}_c}{\text{Precision}_c + \text{Recall}_c},$$

and macro-F1 is

$$F1_{\text{macro}} = \frac{1}{C} \sum_{c=1}^C F1_c, \quad C = 11.$$

Similarly, macro-Precision and macro-Recall are defined as

$$\text{Precision}_{\text{macro}} = \frac{1}{C} \sum_{c=1}^C \text{Precision}_c$$

$$\text{Recall}_{\text{macro}} = \frac{1}{C} \sum_{c=1}^C \text{Recall}_c$$

The following metrics will be reported:

- **Primary metrics:** macro-F1, macro-Precision, and macro-Recall.
- **Secondary metric:** overall accuracy.
- **Error analysis:** confusion matrix.

The test set will not be used to select hyperparameters or checkpoints.

1.9. Baseline Plan

The primary baseline will be a simple CNN (Convolutional Neural Network) trained from scratch without ImageNet pretraining. The model takes an RGB food image of size 512×512 pixels as input and predicts one of the 11 Food-11 food categories.

The baseline consists of an initial convolutional stem followed by four convolutional stages. The stem uses a 3×3 convolution with stride 2, followed by batch normalisation and ReLU activation. Each convolutional stage contains two 3×3 convolutional layers, with batch normalisation and ReLU applied after each convolution, followed by 2×2 max-pooling for spatial downsampling. The channel widths of the four stages are 32, 64, 128, and 256, respectively.

After the final convolutional stage, adaptive global average pooling reduces the feature maps to a single value per channel. A dropout layer with probability 0.3 is then applied before the final linear classification layer, which outputs 11 logits corresponding to the 11 Food-11 classes.

The baseline serves two purposes:

- Establish a reference point for the Food-11 classification task;
- Provide a from-scratch performance reference for comparison with models using pretrained weights and fine-tuning

1.10. Planned Pretrained Models

The main pretrained models will be ResNet-18 and ResNet-50, using the implementations and ImageNet-1K pretrained weights provided by the TorchVision library. The original 1000-class classifier of each model will be replaced with a new 11-class classification head. The experimental design includes two fine-tuning variants for each ResNet architecture:

- **Frozen backbone:** The pretrained feature extractor is frozen and only the new classification head

is trained. This variant is included to evaluate how well the pretrained ImageNet features transfer to Food-11 without updating the backbone parameters;

- **Full fine-tuning:** The complete network is fine-tuned on Food-11 using discriminative learning rates (a lower learning rate for the pretrained backbone and a higher learning rate for the new classification head). This allows the pretrained backbone to adapt gradually to the Food-11 domain while preserving useful ImageNet representations, while the newly initialized classification head can learn the 11-class decision boundaries more rapidly. This variant is included to evaluate whether adapting the pretrained backbone to the Food-11 domain can further improve classification performance.

ResNet-18 and ResNet-50 are included to compare pretrained CNN backbones with different model capacities. ResNet-18 provides a lighter and less computationally demanding backbone, while ResNet-50 provides a deeper and more expressive architecture. Comparing the two models allows the effect of backbone capacity on Food-11 classification performance to be evaluated under the same transfer-learning framework.

1.11. Compute Estimate

The planned experiments will be conducted on a GPU runtime such as an NVIDIA L4-T4 or a comparable device. All main experiments will use the same 512×512 input resolution. The core experiment set consists of one simple CNN baseline trained from scratch and four transfer-learning configurations: ResNet-18 and ResNet-50, each with a frozen-backbone variant and a full fine-tuning variant.

Table 7: Planned computational budget for the main Food-11 experiments.

Item	Planned setting
Main device	NVIDIA L4-T4 GPU
Main resolution	512×512 pixels
Batch size	Initial candidate: 32; reduced to 16, 8, or smaller if required by GPU memory.
Scratch baseline	SimpleCNN: up to 100 epochs, with early stopping based on validation macro-F1
Pretrained models	ResNet-18 and ResNet-50: frozen backbone and full fine-tuning; up to 100 epochs per configuration, with early stopping based on validation macro-F1

Table 8: Measured GPU memory for one forward pass on a batch size 32, 512×512 resolution images

Architecture	Mode	Total params (M)	Trainable params (M)	Before forward (MiB)	Forward peak (MiB)
SimpleCNN	From scratch	1.185323	1.185323	110.42	3,000.42
ResNet-18	Frozen backbone	11.182155	0.005643	148.58	1,172.58
ResNet-18	Full fine-tuning	11.182155	11.182155	148.58	3,680.62
ResNet-50	Frozen backbone	23.530571	0.022539	196.62	1,860.62
ResNet-50	Full fine-tuning	23.530571	23.530571	196.62	13,797.39

The frozen-backbone and full fine-tuning configurations have the same allocated memory before the forward pass. During the forward pass, full fine-tuning requires substantially more memory because intermediate activations are retained for subsequent gradient computation. ResNet-50 full fine-tuning has the highest measured forward-pass peak, at 13,797.39 MiB, compared with 3,680.62 MiB for ResNet-18 full fine-tuning.

These measurements characterize forward-pass memory only. Peak memory during a complete training step must be measured separately, including backward computation, gradients, and optimizer states.

1.12. Subset Selection Rules

The final experiments will use the full Food-11 dataset with its official training, validation, and evaluation partitions.

No class is intentionally excluded and no manually selected subset is planned for the final reported results.